



Coupled 2D–3D CFD approach for innovative simulation of train slipstream in tunnels

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ABSTRACT

The aerodynamic slipstream generated by high-speed trains in tunnels results from the combined effects of the piston wind generated before the train arrival and the local near-train flow. Traditional moving-mesh CFD simulations can reproduce these phenomena with fidelity but at a considerably high computational cost, limiting their use in large parametric or design studies. This work introduces and validates a novel CFD framework that decouples the train–tunnel aerodynamics into two components: a two-dimensional moving-mesh simulation for the piston-wind evolution and a three-dimensional steady-domain simulation with fixed mesh for the near-field slipstream. The methodology is validated against full-scale and reduced-scale reference cases, showing good agreement with experimental data and fully unsteady simulations. The framework also proves its applicability through representative case studies with different tunnel blockages. The proposed approach achieves substantial computational savings, up to two orders of magnitude for the piston-wind estimation and nearly threefold for the near-field simulation, while maintaining physical consistency with the confined flow. It therefore offers an efficient and reliable tool for aerodynamic analysis in railway tunnels, enabling sensitivity and comparative studies that are currently impractical with conventional CFD and supporting engineering and industrial applications.

1. Introduction

The aerodynamic slipstream generated by a moving train is a transient and highly three-dimensional flow phenomenon, characterized by strong velocity fluctuations induced by the train motion through air. The slipstream plays a crucial role in determining aerodynamic loads on nearby structures, objects and passengers, and has therefore been the subject of extensive research in recent years, motivated by safety concerns and regulatory developments. In particular, several incidents involving objects and people on station platforms (e.g. Bowman, 2015, Tages Anzeiger, 2015, RAIB, 2017a, RAIB, 2017b) have highlighted the need for stricter evaluation of aerodynamic effects around trains, especially as train speeds and traffic density continue to increase across Europe and Asia. For these reasons, the assessment of train-induced flow fields, and especially slipstream velocities, is now an essential part of railway aerodynamic design and is explicitly regulated by international standards such as EN-14067 (Standard, 2009a, Standard, 2009b) and the Technical Specifications for Interoperability (TSI, T.S.I. LOC&PAS, 2014).

While most previous and current investigations have focused on open-air conditions, the aerodynamic problem becomes significantly more complex in confined environments, such as underground tunnels

and station platforms, where the surrounding walls strongly modify the flow field. In these conditions, the train entry and motion induces a piston effect, a bulk displacement of air along the tunnel ahead of the train; at the same time, a reverse flow develops in the annular gap between the train and the tunnel walls, generated by the combined influence of tunnel blockage, the local reduction of the available cross-sectional area caused by the train presence and the pressure difference between the train nose and tail, which superimposes on the local near-field slipstream. The resulting confined flow is therefore governed by a strong coupling between train and tunnel induced aerodynamics, depending on the train induced flow, tunnel geometry, blockage ratio, and longitudinal pressure gradients.

Experimental studies in confined environments are relatively limited due to the inherent difficulties in reproducing large-scale tunnel conditions. Full-scale investigations conducted in underground railway tunnels, such as the dedicated experimental campaign on the Italian line of Negri et al. (2023, 2025c), or the measurements from the Chinese Xikema tunnel, later used for CFD validation (e.g. Jiang et al., 2019, Du et al., 2024), have provided valuable reference data, although they remain expensive and closely tied to the specific conditions of each test site.

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Scaled-model tests, such as those carried out at the TRAIN Rig facility using the ICE2 model within different partially-closed or closed infrastructure (Gilbert et al., 2013), or the work of Meng et al. (2019) to assess the slipstream caused by a metro train passing through a tunnel, offer controlled conditions and repeatability, but are constrained by Reynolds scaling effects and limited tunnel length and overall dimensions.

As a result, Computational Fluid Dynamics has become the primary tool for investigating train-induced flows in tunnels, enabling full access to the flow field and detailed aerodynamic characterization. Most numerical studies employ moving-mesh simulations, where the train motion is explicitly modeled within a large computational domain representing the tunnel and adjacent open air regions. This approach enables the accurate reproduction of transient aerodynamic effects associated with the train entry, motion, and exit, allowing detailed analysis of flow development and pressure wave propagation. Several studies, employing this methodology, have focused on the aerodynamic effects directly associated with the train configuration and motion itself. Jiang et al. (2019) analyzed train-related effects by simulating high-speed trains with different marshalling configurations entering a tunnel, while Li et al. (2025c) investigated active flow-control strategies aimed at reducing aerodynamic loads during tunnel entry. Additionally, Iliadis et al. (2020) analyzed a scaled freight train model to evaluate the aerodynamic field in a simplified single-track tunnel setup, comparing the effects of high-speed and freight train configurations, while Li et al. (2025a) examined the aerodynamic interaction between crossing trains in tunnels. The influence of tunnel geometry has been addressed as well: Li et al. (2025b) examined tunnels with variable cross sections and Xiong et al. (2022) studied the aerodynamic behavior of a high-speed train running through a tunnel including a lateral cave. Other works, such as Fu et al. (2017), evaluated the flow in double-track tunnels, highlighting the asymmetric flow development between the two sides and the sensitivity to tunnel length. The methodology has also been extended to non-conventional train systems, including maglev trains, as in Liu et al. (2023), where the unsteady aerodynamic field was characterized for different train nose lengths. From a more detailed aerodynamic safety standpoint, Khayrullina et al. (2015) performed large-eddy simulations to analyze the slipstream generated on station platforms, representing one of the few studies explicitly focused on confined environments with platforms. These studies demonstrate the capability of moving-mesh CFD simulations to reproduce complex aerodynamic phenomena in tunnels, capturing the coupled effects of confinement, train geometry, and tunnel dimension and shape on pressure and velocity fields. The moving train approach, although physically accurate, involves very high computational costs because it requires simulating long transients covering the entire train entry, passage, and exit through the tunnel. The need for large domains further increases the number of elements and the expense of mesh motion and interpolation, resulting in long computation times. Such high costs restrict the possibility of performing a large number of simulations, limiting the exploration of different train configurations and operating conditions. While the approach is suitable for detailed analyses or validation studies, it is not practical for broad sensitivity or parametric investigations. Recent advances have also explored more efficient alternatives to moving-mesh simulations: Ouyang et al. (2025) employed a wall-modeled LES approach with a stationary train, validating the results against wind-tunnel data. Although this setup allowed a reduction in computational cost, it lacked realistic boundary conditions representative of a moving train within a confined domain and was therefore limited to reproducing the experimental configuration and study specific bottom flow patterns, rather than the full tunnel–train aerodynamic interaction.

Finally, there is still no efficient and reliable methodology capable of reproducing both the pre-existing tunnel flow and piston wind conditions in confined spaces and the near-field slipstream dynamics within reasonable computational times, which strongly limits the possibility of

parametric or design sensitivity analyses. The present work addresses these gaps by introducing a novel and computationally efficient CFD framework for train–tunnel aerodynamics. The methodology separates the piston-wind field ahead of the train from the local near-train aerodynamics, enabling a physically consistent yet efficient representation of the confined aerodynamic environment. The piston-wind component is modeled through a two-dimensional moving-mesh simulation, while the near-train region is reproduced using a three-dimensional steady-domain simulation with fixed mesh, where the inlet boundary conditions are tuned according to the piston-wind velocity extracted from the 2D analysis. This coupled approach drastically reduces computational cost while maintaining consistency with the governing flow physics. A further key advantage of the proposed framework is that it enables the extraction of slipstream statistics, i.e. average and standard deviation velocity profiles, allowing the method to reproduce not only the instantaneous flow features but also its statistical behavior, which is essential for slipstream assessment. This capability is verified by comparing the flow statistics obtained from the proposed framework with those derived from full moving mesh simulations, using the ensemble based validation procedure presented in Negri et al. (2025b). Finally, beyond its methodological novelty, the proposed framework is conceived as a practical tool for engineering applications. By reducing computational requirements while retaining physical reliability, it enables efficient qualitative and quantitative analyses for a wide range of train and tunnel configurations, allowing for sensitivity assessments and preliminary design studies that are currently impractical with conventional CFD simulations, due to excessive computational cost.

The paper is organized as follows. Section 2 introduces the conceptual framework and the reference geometries for train and tunnel configurations. Section 3 presents the numerical methodology, including the two-dimensional piston-wind estimation (Section 3.1) and the three-dimensional near-field flow modeling (Section 3.2), together with their validation and verification. Section 4 illustrates the application of the methodology to representative tunnel–train configurations with different blockage ratios. Finally, Section 5 summarizes the main findings and outlines future developments.

2. Conceptual framework

This section outlines the framework developed to simulate train-induced aerodynamic flows in tunnels, describing the conceptual approach and the reference geometries adopted for both full-scale and scaled train configurations.

2.1. Physical modeling approach

When a train enters and travels inside a tunnel, two distinct aerodynamic flow regions can be identified at a fixed measurement point along the tunnel: the piston wind region, established prior to the arrival of the train at the measurement location, and the near-field slipstream region.

From the moment the train nose enters the tunnel until the train passes the measurement point, the flow is predominantly one-dimensional and is commonly modeled using one-dimensional or simplified two-dimensional approaches, as in Pan et al. (2013), Yan et al. (2020), Liu et al. (2018). The blockage effect of the train, combined with its aerodynamic drag, induces a bulk flow along the entire tunnel: air is pushed forward in the direction of train motion. Under real conditions, air compressibility is inherently present and gives rise to pressure waves (train–tunnel pressure signature), which travel along the tunnel and interact with portals and local cross-sectional variations (e.g. narrowings, niches). In the present study, however, the simulations are performed under incompressible flow assumptions. Consequently, the propagation of compressible pressure waves is not reproduced, and the flow field is solely governed by the displacement and drag of the train body. This modeling choice allows to focus on the

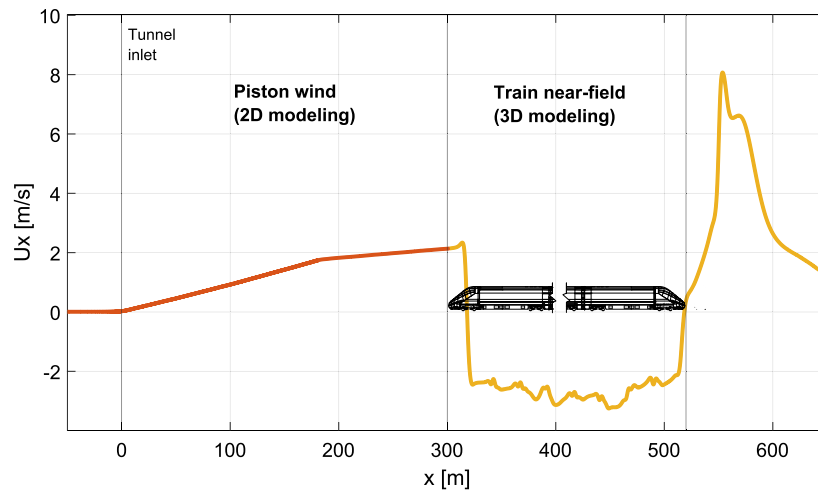


Fig. 1. Qualitative illustration of the longitudinal air velocity evolution induced by a high-speed train in a tunnel, obtained from a CFD simulation of the configuration whose geometric and operating parameters are detailed in Section 2.2.1. The piston wind region is highlighted in orange, while the near-field slipstream and wake regions are shown in yellow.

convective piston effect without the superposition of compressibility-driven transients. Within this region, the velocity measured in the tunnel, as shown in Fig. 1, exhibits an initial approximately linear increase during the entrance of the train nose and body. A second, more moderate slope follows as the train fully enters the tunnel. This reduced velocity growth rate can be attributed to the pressure field induced by the train wake, which lowers the intensity of the piston flow.

Once the train passes in front of the measurement point, the flow is dominated by the unsteady boundary-layer development and the wake structures generated by the train. The near-field slipstream signature exhibits a well-defined temporal evolution that can be described in three successive stages. As the nose of the train passes, the velocity signal shows a sharp reversal: the piston wind contribution rapidly decreases from positive to negative values, in the direction opposite to the train motion. This behavior arises from the local blockage between the train cross-section and the tunnel wall, combined with the pressure difference between the train nose and the rear region. During the subsequent passage of the train body, the boundary layer developing along the surface grows and interacts with the flow within the annular gap between the train and the tunnel wall. Finally, as the tail of the train crosses the measurement position, the velocity undergoes an abrupt transition back to positive values. This is followed by a pronounced velocity peak, typical of high-speed trains, caused by the pair of counter-rotating vortices shed from the tail. These vortical structures dominate the near-wake, for this train type, and are responsible for the maximum slipstream intensities observed in open air and inside tunnels as well.

The two flow regimes are driven by different physical mechanisms and, in the present work, they are treated separately, with the aim of modeling the piston wind as a predominantly one-dimensional bulk flow and the near-field slipstream as a highly three-dimensional unsteady wake. No linear superposition is assumed between the near-field slipstream that the train would generate in undisturbed conditions and the piston effect developing in the annular region, as the resulting counter-flow modifies the boundary-layer development on the train surface through altered shear conditions. The only superimposed contribution considered is the piston wind established prior to train arrival, while the flow field in the train vicinity arises from the combined interaction between near-field slipstream and annular piston effect, which are simultaneously resolved by the 3D numerical model.

The decoupling process described herein, schematically illustrated in Fig. 2, is introduced as a working hypothesis: by separating these two contributions, reproducing each component with dedicated models, it becomes possible to represent the unsteady slipstream in terms of its

time-averaged or instantaneous behavior, allowing for a more tractable yet physically consistent representation of the complex train-tunnel aerodynamics.

2.2. Validation reference geometries

The reference train and tunnel geometries are described here, followed by a summary table (Table 1) reporting the role of each configuration and data source in the present study.

2.2.1. Full-scale ETR1000 train in underground tunnel

The first validation activity focuses on a full-scale high-speed passenger train operating in an underground railway tunnel inspired by the real configuration of the Turin Rebaudengo line in Italy, but adopting an idealized geometric representation. The Rebaudengo case represents a real scenario in which the aerodynamic effects of interest, namely the piston wind and the near-field slipstream, can be directly observed at full scale and compared with available data from a previous experimental campaign, whose setup and related results are extensively described in Negri et al. (2023) and Negri et al. (2025c). However, the main limitation of this real-case tunnel lies in its varying cross-sectional area, which prevents consistent sampling of the slipstream velocity at multiple, regularly spaced locations along the platform and ultimately precludes the extraction of independent measurements required for an ensemble-based slipstream statistics (Negri et al., 2025b). This is particularly critical because the slipstream is an inherently stochastic phenomenon; a single train passage provides only a single realization of the flow field and cannot capture the statistical variability required for a robust characterization. For this reason, the experimental configuration is not used to directly derive the results of the present analysis, but rather to develop an alternative idealized scenario and inform the development of the numerical model adopted in this work.

To overcome the experimental data limitation, an idealized tunnel model has been created and employed in this work starting from the Turin Rebaudengo tunnel geometry, preserving a comparable length of the real tunnel section of interest (600 m) but adopting a uniform cross-section along the entire tunnel (Fig. 3(b)), while maintaining a blockage of 13%. Although such geometry does not correspond to an existing infrastructure, it is deemed to be realistic and suitable to enable the application of the methodology proposed in Negri et al. (2025b), which allows an ensemble-averaged slipstream representation to be extracted from a single CFD simulation.

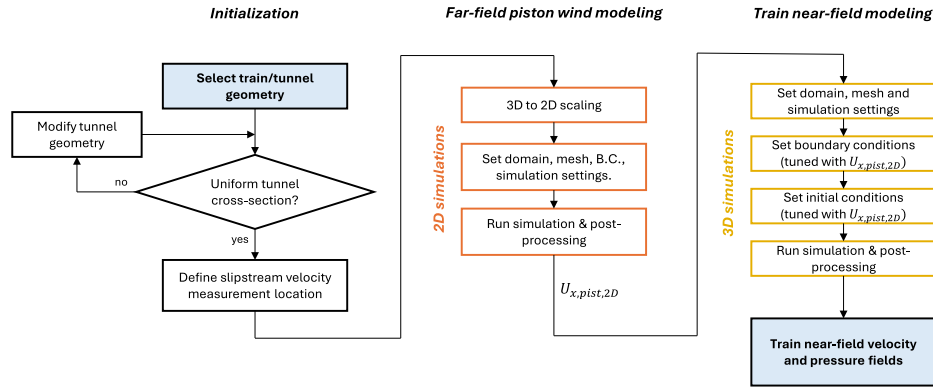


Fig. 2. Block diagram of the proposed decoupling framework.

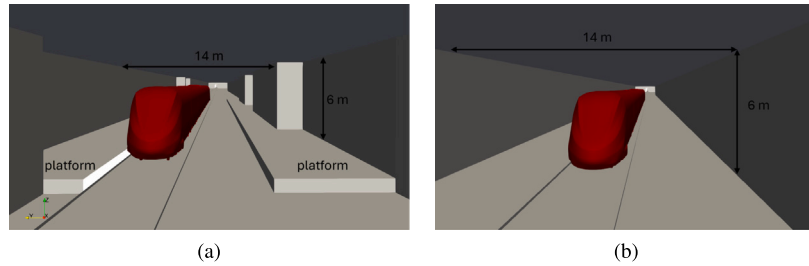


Fig. 3. Underground tunnel configurations used for model validation. (a) Turin Rebaudengo tunnel with a non-uniform cross-section at the platform region. (b) Idealized uniform-section tunnel derived from the Turin geometry, preserving the same overall length and blockage but with uniform cross-sectional area.

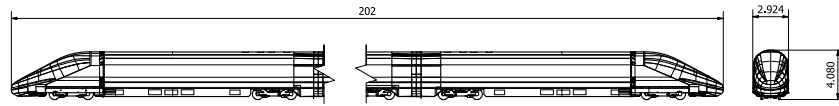


Fig. 4. ETR1000 high-speed train geometry details and overall dimensions in meters.

The train considered corresponds to the ETR1000 (Fig. 4), a 202 m long high-speed passenger train composed of eight cars in fixed configuration. Its operating speed in the tunnel is set to 120 km/h (33.3 m/s), consistent with the average passing speed observed on the real line.

2.2.2. Scaled-model ICE2 train in TRAIN Rig facility

A second reference configuration is considered through a reduced-scale high-speed train model tested in a controlled laboratory environment. The train corresponds to a 1:25 scale representation of the German ICE2 high-speed passenger train (Fig. 5) traveling at about 115 km/h (32 m/s), experimentally investigated within the TRAIN Rig facility at the University of Birmingham (Gilbert et al., 2013) and numerically assessed in Negri et al. (2025b).

The TRAIN Rig features a single-track rectangular tunnel with constant cross-section along its length. In the experimental campaign of Gilbert et al. (2013), the tunnel measured approximately 7.96 m in length, with an internal height of 0.279 m and width of 0.262 m. The corresponding train–tunnel blockage ratio is approximately 23%. The scale-model consists of four coaches including inter-carriage gaps and simplified bogies, with an overall length of 4.3 m and a frontal area of 0.017 m². In addition to the original experimental tunnel configuration (Gilbert et al., 2013), a longer tunnel with the same cross-section but extended length has also been numerically investigated in Negri et al. (2025b). This extended configuration (tunnel length = 24 m) allows the slipstream to evolve further downstream of the entrance region, enabling the extraction of multiple, statistically independent velocity samples from a single train passage, and is therefore used in the present work to support ensemble-based analysis of the near-field flow.

The inclusion of this scaled-model reference case serves two purposes: it provides an additional validation of the simplified modeling strategy under different blockage levels, and Reynolds number conditions, and it enables a further comparison against documented experimental observations obtained within a highly controlled environment.

3. Numerical methodology and models verification

3.1. Piston wind estimation through 2D simulations

The first modeling approach is introduced to represent the piston wind phenomenon, based on a two-dimensional CFD framework with a moving-mesh strategy. This model is employed to reproduce the initial transient phase that occurs as the train enters the tunnel and generates the bulk air displacement characteristic of the piston effect. A two-dimensional numerical model has been chosen in order to maintain a fully CFD-based environment while limiting the computational cost compared to full three-dimensional simulations. This approach requires only geometrical inputs, i.e. the external shape of the train and the cross-sectional layout of the tunnel, without the need for empirical coefficients or parameter estimation. In this way, the piston wind can be evaluated directly from the train–tunnel configuration, using the same type of numerical framework adopted for the subsequent near-field slipstream analysis, while avoiding the introduction of additional modeling assumptions.

3.1.1. Methodology and geometrical parameters

The numerical methodology adopted to reproduce the piston wind in a two-dimensional domain builds upon the approach proposed by Liu



Fig. 5. ICE2 scaled-model geometry and overall dimensions in meters (1:25 scale) (Gilbert et al., 2013).

Table 1

Reference configurations and validation scope adopted in the present study.

ETR1000 (full scale)			
Data source	Numerical model validation		Overall methodology validation
	2D CFD	3D CFD	
Numerical CFD data	✓		✓
Experimental data (full scale)		✓	
ICE2 (1:25 scale)			
Data source	Numerical model validation		Overall methodology validation
	2D CFD	3D CFD	
Numerical CFD data	✓		✓
Experimental data (scaled model)	✓	✓	

et al. (2018), who demonstrated that the three-dimensional train-tunnel flow field can be simplified into a dynamic 2D model without compromising the main aerodynamic features. Their work systematically compared several 2D formulations against both experimental measurements and full 3D simulations for train-tunnel configurations, showing that a properly scaled 2D representation can accurately capture the transient bulk flow induced by a moving train. In particular, the methodology relies on reducing the real 3D tunnel cross-section to an equivalent 2D channel by preserving the blockage ratio (β) and defining an effective tunnel width through the hydraulic diameter of the original cross-section, while the train width is adjusted to maintain the same volumetric occupation.

In the present study, the 2D computational domain (Fig. 6) is constructed by considering the key geometric quantities of the real configuration. The tunnel is modeled as a straight channel of length L_{tunnel} and equivalent width W_{tunnel} . To account for the actual wetted perimeter and the associated wall-friction effects, the equivalent tunnel width is defined through the hydraulic diameter of the original 3D cross-section, given by Eq. (1), where A_{tunnel} is the tunnel cross-sectional area and P_{tunnel} its wetted perimeter.

$$W_{\text{tunnel}} = D_h = \frac{4 A_{\text{tunnel}}}{P_{\text{tunnel}}} \quad (1)$$

Once the tunnel width is defined, the equivalent train width W_{train} is determined by preserving the blockage ratio. The train is therefore represented with its original length L_{train} and an adjusted width W_{train} , while the tunnel width is fixed to its hydraulic diameter D_h . This scaling preserves the key physical effects governing the piston wind: it reproduces the correct longitudinal acceleration of the air induced by the train's displacement and maintains the appropriate wall-shear interaction and frictional losses, as typically done in internal flow analyses. A practical adjustment is required to implement this rescaling in the numerical model: while the tunnel geometry can be directly defined with the desired equivalent two-dimensional width, the train geometry must be adapted starting from the real three-dimensional model. In this work, the train is modified by applying a uniform scaling factor in the lateral direction so that its width matches the value calculated from the blockage ratio relation. This transformation slightly distorts the original train shape but, as it will be shown in the results, it appears to have a minor influence on the predicted piston wind. Once the correct lateral dimension is obtained, the 3D train model is converted to 2D by extracting a single horizontal section. The cut is performed at the height corresponding to the maximum width of the train nose, so that a representative frontal shape is retained. Although this simplification is considered adequate for the scope of this study, it may introduce some differences for trains with highly three-dimensional nose shapes, such as modern high-speed trains.

3.1.2. Numerical setup, computational domain and boundary conditions

The two-dimensional simulations are carried out using an unsteady Reynolds-averaged Navier-Stokes (URANS) formulation with the k- ω SST turbulence model, which has proven effective for capturing the aerodynamic behavior of trains (Menter, 1993, Menter, 1994, Hemida, 2023). The numerical solution is advanced with the PIMPLE algorithm. Incompressible flow is assumed in order to isolate the drag-induced bulk motion of air from the pressure-wave phenomena that would arise in a compressible regime. This assumption, while neglecting wave propagation, allows the present work to focus on the modeling strategy for the induced flow and to develop a methodology that can later be extended to compressible conditions without mixing the two effects in the current validation stage. The approximation is considered reasonable for the validation cases examined here, including the full-scale uniform tunnel, where the average blockage ratio is only about 0.13, the train speed is moderate, and the compressibility and pressure waves effects to the overall flow field are limited; in the reduced-scale TRAIN Rig tests with the ICE2 model, the tunnel is short enough that pressure waves propagate, reflect, and dissipate rapidly, producing only small, transient disturbances with limited amplitude in the local air velocity. The motion of the train inside the tunnel is reproduced through a sliding-mesh technique, a well-established approach for problems with interacting moving components. This method has been widely adopted in aerodynamic studies of tunnels and other confined or partially confined environments to evaluate vehicle stability Khayrullina et al. (2015), Brambilla et al. (2021), Araya Reyes et al. (2022), Semeraro et al. (2022) and is the same strategy employed for the case study of the ETR1000 train in the Turin Rebaudengo tunnel (Negri et al., 2024, Negri et al., 2025a). The computational domain is divided into a stationary region, representing the tunnel walls and external surroundings, and a moving region representing the train. An interface boundary couples the two regions and transfers the flow solution during relative motion. Spatial discretization is performed with the finite-volume method as implemented in the OpenFOAM solver. The domain includes both the open-air entry section and the tunnel configuration shown in Fig. 6, with upstream and downstream dimensions selected to minimize far-field interference in the open-air portions. Considering the boundary conditions adopted in the simulations, no external inflow velocity is prescribed; the entire airflow is generated by the motion of the train as it travels from the open-air section into the tunnel. The initial turbulence intensity, used to initialize the turbulence model parameters, is set to 2%. The tunnel walls are modeled with a no-slip boundary condition. At the upstream end of the open-air region, the velocity is fixed to zero, while the downstream boundary is defined as a pressure outlet with zero relative static pressure. The lateral patches enclosing the open-air portion are also treated as outlets, both upstream

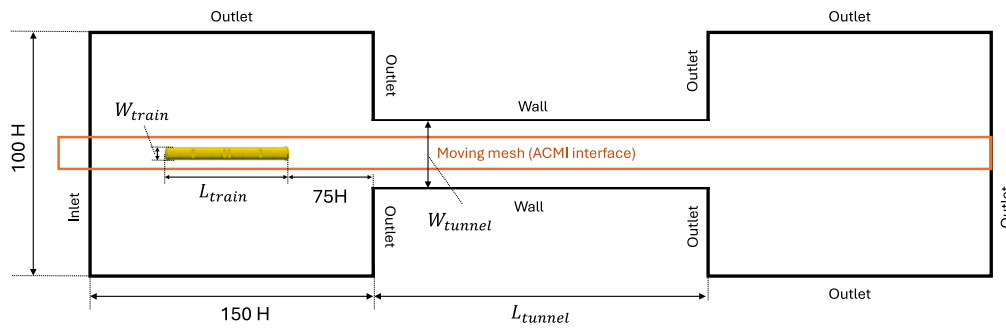


Fig. 6. Scheme of the 2D numerical domain with dimensions and boundary conditions. The moving mesh region is shown in orange, while the stationary domain is shown in black. H represents the train height, taken as reference dimension.

and downstream of the tunnel entrance, to allow the entrainment and discharge of air induced by the train motion. The transition between the stationary tunnel mesh and the moving mesh surrounding the train is handled by the Arbitrary Coupled Mesh Interface (ACMI), which provides velocity, pressure, and turbulence fields data exchange across adjacent but non-conformal mesh regions. The train surface, included in the moving mesh, is treated with an adherence (no-slip) condition. Its velocity is imposed through the movingWallVelocity boundary type, which prescribes the instantaneous wall velocity consistent with the sliding mesh motion. As the train itself generates its velocity profile, no explicit initialization of velocity was made in the domain (a velocity reference of about 0 m/s was given to the inlet region). The train was positioned at a normalized distance of $75H$ from the tunnel inlet, to allow for flow speed stabilization due to numerical transients, where H represents the train height, taken as reference dimension. For the near-wall treatment, standard wall functions are applied on both the train surface and the tunnel walls. This approach provides an accurate representation of the boundary layer development while maintaining numerical robustness and computational efficiency. Spatial discretization of the convective and diffusive terms employs a second-order central difference scheme, whereas time advancement is carried out using the second-order implicit Crank–Nicolson method. The time step is selected to satisfy a strict Courant–Friedrichs–Lewy (CFL) stability condition, with the maximum CFL number kept below 1, resulting in an average time step of 1 ms. This choice is dictated by the use of the PISO algorithm, which treats advection terms explicitly without outer iterations and therefore requires a sufficiently small Courant number to preserve temporal accuracy. This criterion inherently reflects the smallest cell sizes within the computational mesh and guarantees a temporal resolution adequate to capture the unsteady flow structures of interest. Under this approach, the time step is directly linked to the local convergence behavior.

3.1.3. Mesh

The mesh analysis primarily focuses on the full-scale ETR1000 configuration, which is selected as the main reference case for assessing mesh quality and verifying the 2D numerical model. The same meshing strategy and spatial resolution are later applied to the scaled-model ICE2 configuration, where the computational domain and mesh are geometrically scaled according to the model dimensions.

The numerical discretization is generated in OpenFOAM using the snappyHexMesh utility, which enables local refinement in selected regions of the computational domain, including the train surface, the tunnel interior, and the surrounding open-air sections. Within the tunnel, particular care is taken to maintain comparable cell sizes across the stationary and moving mesh regions to ensure smooth interpolation at their interface. Additional refinement is applied in the wake region of the moving mesh to capture the train-induced pressure drop and overall aerodynamic drag. While the model is two-dimensional and does not reproduce the full three-dimensional flow structures, this refinement

strategy ensures an accurate representation of the pressure field and the drag-driven piston wind effects relevant for the present study.

To model the boundary layer, prism layers were inflated on the train surfaces. Two layers setup were tested on the train surface, adding 5 or 10 layers. The transition between the layers and the outer hexahedral mesh was handled by selecting an expansion ratio of 1.2 for the prism layers. Fig. 7 illustrates the mesh structure in tunnel and train near regions.

To verify that the numerical results are independent of the mesh resolution, four different grids were tested and the corresponding piston-wind velocity profiles along the tunnel are compared in Fig. 8. Each mesh, defined as coarse, medium and fine, shares the same overall topology and refinement regions around the train and within the domain, while only the base cell size is varied and the refinement levels are scaled accordingly. This strategy enables a systematic assessment of mesh sensitivity while keeping the mesh structure consistent. An additional configuration, referred to as medium V2, was generated by modifying only the number of prism layers on the train surface from 5 to 10 in order to evaluate the effect of near-wall resolution, particularly on y^+ values and the resulting velocity profiles. The results show that all meshes except the coarse one lead to nearly identical piston-wind predictions. This indicates that, for the present two-dimensional configuration, values of y^+ above 300 do not significantly affect the solution, since the medium and medium V2 meshes provide almost indistinguishable results. Furthermore, the medium meshes achieve the same velocity solution as the fine grid while using only about one third of its total number of elements. Based on this analysis, the medium mesh with ten near-wall layers is selected as the reference grid for the subsequent simulations. Although the medium mesh with five layers also performed satisfactorily, the version with ten layers was preferred to maintain y^+ values within the commonly accepted range, despite the limited sensitivity observed in this specific case.

3.1.4. Model verification

The proposed two-dimensional piston wind model was verified by comparing its predictions against both full-scale and reduced-scale reference cases in order to test its robustness under different geometric configurations and Reynolds number conditions. The validation results are presented in Fig. 9.

For the ETR1000 full-scale configuration, which results are shown in Fig. 9(a), the validation consists of a direct comparison between the present 2D model and a fully three-dimensional moving-mesh simulation performed on the equivalent tunnel. The 2D solution well reproduces the piston wind profile during the train entry phase (from 0 to approximately 202 m), with the velocity gradient matching that obtained from the 3D simulation. In the second part of the signal, once the train is fully inserted into the tunnel (beyond 202 m), the 2D model maintains a comparable, but slightly overestimated, slope to the 3D reference, overestimating the final flow velocity of about 0.1 m/s (relative error of 2.8%).

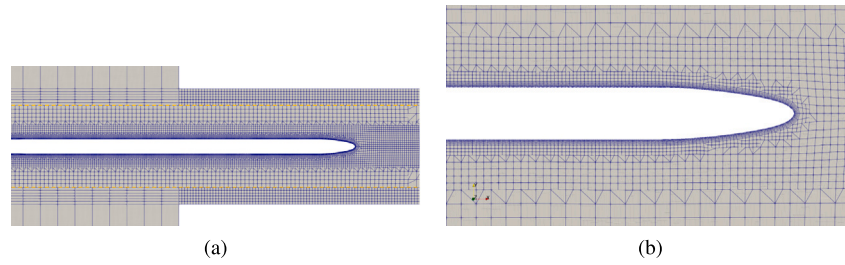


Fig. 7. Details of the 2D model mesh: (a) mesh structure at the tunnel inlet; (b) mesh refinement in the vicinity of the train nose.

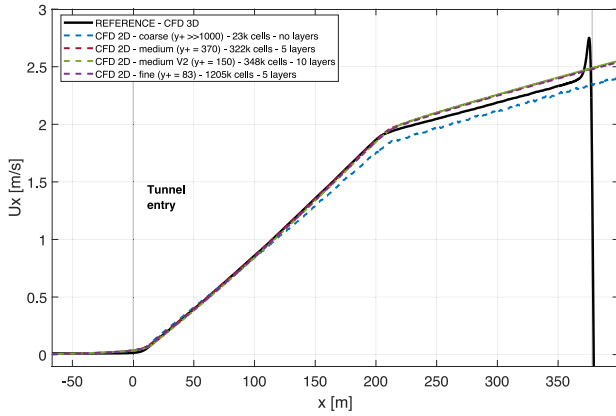


Fig. 8. Mesh independence analysis for the 2D model. Longitudinal piston wind velocity is compared across four mesh resolutions, with the 3D simulation result shown for reference (black line).

The second validation concerns the reduced-scale ICE2 train tested in the TRAIN Rig facility at a 1:25 scale. The results are compared with experimental measurements reported by Gilbert et al. (2013) and with the 3D moving-mesh simulations presented in Negri et al. (2025b), which are also included in Fig. 9(b). In this case, the measurement point

is reached by the train at about 5.5 m, when the vehicle is already fully inside the tunnel but has only just entered. The 2D model still reproduces the piston-wind profile with high accuracy, closely matching both the experimental data and the 3D CFD results. A validation is also conducted for the extended-length tunnel configuration analyzed in Negri et al. (2025b) (Fig. 9(c)), where the 2D model remains in reasonable agreement with the 3D reference. A deviation appears once the train is fully inside the tunnel, with a maximum relative error of about 8% at the end of the piston wind ramp. This behavior is likely linked to the geometric simplification introduced when converting the three-dimensional train into a two-dimensional representation by cutting a longitudinal section at the nose. Since the ICE2 nose exhibits pronounced three-dimensional curvature, this reduction seems to alter the induced drag distribution, leading to an overestimation of the wind speed-up in the 2D model.

Overall, the comparisons confirm that the proposed 2D methodology provides an accurate representation of the piston wind across very different conditions, including full-scale tunnels with moderate blockage ratios and reduced-scale facilities with higher blockage ratio and a significantly different Reynolds number. The main limitation observed is a slight overestimation of final piston flow, possibly linked to the geometric simplifications inherent in reducing a three-dimensional nose to a two-dimensional section. Nonetheless, the discrepancy is considered minor and is not expected to affect the practical applicability of the method.

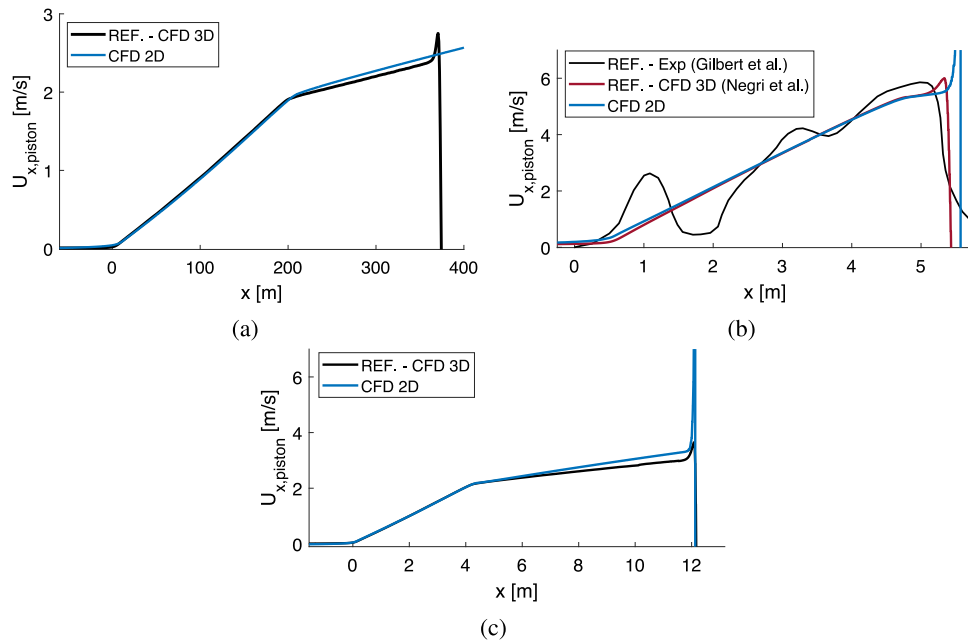


Fig. 9. Longitudinal piston wind velocity profiles for model verification: (a) ETR1000 high-speed train in the uniform-section tunnel; (b) ICE2 train at 1:25 scale in the TRAIN Rig (tunnel length = 7.9 m); (c) ICE2 train at 1:25 scale in the TRAIN Rig (tunnel length = 24 m).

3.1.5. Computational efficiency and limitations

In addition to requiring no input parameters beyond the geometric definitions of the train and tunnel, which would be needed in predictive analytical models, the proposed two-dimensional approach also provides a significant advantage in terms of computational efficiency compared with full three-dimensional simulations. For the full-scale ETR1000 train, the 2D grid contained about 343 thousand cells, while the corresponding 3D model required more than 11 million cells, meaning that the two-dimensional mesh size was roughly 3% of the three-dimensional one. To simulate a 10s physical time window under the same HPC environment, the 3D simulation required 5174 core hours, whereas the 2D simulation was completed in only 22 core hours. Defining the computational speed up as the ratio between the 3D and 2D core hours, the two-dimensional model achieved a speed up of more than 230, over two orders of magnitude faster than the corresponding 3D computation. This dramatic reduction in cost makes it possible to determine the piston wind field very efficiently and to use it as a boundary condition for subsequent 3D slipstream analyses, avoiding the need to resolve this part of the flow in full three dimensions.

Beyond its computational performance, the 2D simulation is straightforward to set up, requires no special calibration and only needs the geometric definition of the train and the tunnel. This makes it practical and robust for engineering applications, where short development time and limited input data are often critical. It is acknowledged that an even simpler one-dimensional analytical model (e.g. Liu et al., 2020, Yan et al., 2020) could provide piston wind predictions in a shorter time; however, such models usually require case specific tuning or empirical coefficients that depend on the train and tunnel configuration. By contrast, the proposed 2D CFD approach works entirely within the same numerical framework used for the subsequent 3D analysis and does not rely on additional fitting or external parameters, representing a reasonable and practical alternative to purely analytical approaches, and is therefore employed in the present study.

3.2. Near-field flow estimation through 3D simulations

Three-dimensional simulations are performed to reproduce the aerodynamic flow field when the train is fully immersed in the tunnel, with the piston wind already developed and no additional transients generated by the train entering from open air or local tunnel geometry modifications. The main objective is to extract the near-field aerodynamic quantities around the train under quasi-steady operating conditions, representative of a train traveling at constant speed inside a long underground section. In this configuration, the non-stationarity of the flow is entirely due to the train aerodynamics in a confined environment, while effects associated with the initial entrance of the train into the tunnel or with local geometric discontinuities are intentionally excluded. With this aim, the computational domain is designed as a tunnel of uniform cross section, which can represent either a simple tunnel without platform or a tunnel with a platform region assumed to be sufficiently long to avoid end effects and therefore not influenced by boundary-related flow disturbances. This assumption justifies the use of a geometrically uniform domain, avoiding the influence of local variations in tunnel shape or platform geometry on the simulated flow. The analysis aims to determine whether the steady-domain methodology here proposed can reproduce the same flow statistics, as a full moving-mesh simulation, by comparing the ensemble averages and standard deviations obtained with the ensemble-based approach (Negri et al., 2025b) to the time-averaged and fluctuating quantities predicted by the static-mesh model.

The three-dimensional model analysis is performed, as for the 2D modeling, for the two reference configurations of the full-scale ETR1000 train operating in a uniform-section underground tunnel, and for the scaled ICE2 train model tested within the TRAIN Rig tunnel.

3.2.1. Numerical setup, computational domain and boundary conditions

The three-dimensional simulations are performed using an unsteady Reynolds-averaged Navier–Stokes (URANS) formulation with the $k-\omega$ SST turbulence model, advanced in time with the PIMPLE algorithm. An incompressible flow assumption is employed, which is considered reasonable for the validation cases examined, as commented in Section 3.1.2. The flow field around the train is computed using a static mesh, while the boundary conditions are adjusted to reproduce the effective relative velocity that the train would experience once the piston wind is developed inside the tunnel. A key aspect of this setup is the prescription of the inlet velocity. Because the train itself generates the piston wind, the flow entering the computational domain must represent the relative air speed acting on the train once this bulk motion is established. To achieve this equivalence, the train is kept fixed with a no-slip condition applied to its surface, while the tunnel walls are assigned a velocity equal to the negative of the train's operating speed. At the inlet, the imposed air velocity is set to the difference between the train speed and the piston-wind velocity predicted by the two-dimensional simulations, sampled when the train nose reaches a predefined reference position inside the tunnel, sufficiently far from the entrance to ensure a quasi-stationary piston-wind regime (the specific definition for each case study is reported in Section 3.2.4). The imposed inlet velocity is prescribed as uniform across the tunnel cross-section, consistent with the predominantly bulk-flow nature of the piston wind, while near-wall velocity gradients develop within the domain through the relative velocity boundary conditions applied at the tunnel walls.

The outlet is defined as a pressure outlet at zero gauge pressure. The same relative velocity used at the inlet is also applied as the initial condition throughout the air volume. The initial turbulence intensity, used to initialize the turbulence model, is set to 2%.

Spatial discretization is carried out using the finite-volume method implemented in OpenFOAM. The computational domain consists of a tunnel with a uniform cross section along its entire length. To avoid inlet and outlet interference, the train is positioned 25H (H: train height) upstream of the inlet and 150H away from the downstream boundary, allowing for wake development and dissipation. Upstream and downstream extents are selected to minimize boundary effects from Brambilla et al. (2022), where comparable domain dimensions were adopted for train–tunnel aerodynamic studies.

Standard wall functions are applied on both the train surface and the tunnel walls for near-wall modeling. The numerical discretization schemes follow the same strategy used in the two-dimensional simulations. Convective and diffusive terms are approximated using a second-order central difference scheme, while temporal integration is performed with an implicit Crank–Nicolson method. The time step is chosen to satisfy a strict Courant condition, keeping the maximum CFL number below 2 to ensure stability and adequate temporal resolution. The higher CFL limit compared to the 2D simulations is enabled by the use of the PIMPLE algorithm, which employs outer iterations allowing accurate transient computations at Courant numbers greater than unity. This choice also preserves consistency with the temporal discretization adopted for the moving-mesh simulations employed as the benchmark for verifying the present model.

The overall boundary condition setup and domain layout, common to both the ETR1000 and ICE2 cases, are illustrated in Fig. 10.

3.2.2. Mesh

The three-dimensional computational mesh is generated using snapPyHexMesh, with Fig. 11 showing a representative excerpt of the discretization used for the ETR1000 case. Sub-figure (a) highlights the upstream region in front of the train, where a sequence of refinement boxes is employed to adequately resolve the incoming flow before impingement on the nose. The innermost refinement region extends approximately 11 m upstream of the train nose, while a second, coarser refinement level extends up to 36 m. In the downstream wake region, the finest refinement box extends 50 m behind the train tail to capture

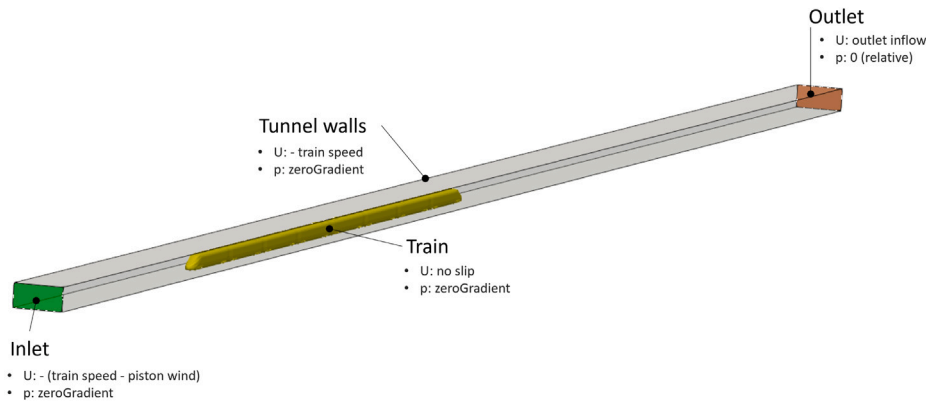


Fig. 10. Boundary conditions setup for 3D stationary model.

the near-wake structures, followed by a coarser refinement region extending up to 150 m to allow wake dissipation without boundary interference. Around the train surface, near-wall layers is applied using 10 prism layers with a growth rate of 1.35, designed to maintain the non-dimensional wall distance within the operating range of the wall functions adopted by the turbulence model ($30 < y^+ < 300$). Outside these regions, the mesh resolution transitions progressively through distance-based refinements in order to limit excessive cell clustering while preserving accuracy.

The detailed mesh specification adopted for the full-scale ETR1000 case, including mesh size selection, grid-independence assessment, and overall discretization strategy, has been documented in Negri et al. (2024, 2025a), which present the slipstream numerical analysis conducted in the Turin Rebaudengo tunnel. The same mesh setup is consistently employed in the present work, and only the mesh features and information essential for the simplified methodology are outlined here, while full details can be found in the cited works.

For the ICE2 scaled-model simulations, the meshing strategy and numerical setup adopted are extensively reported in Negri et al. (2025b), therefore, they are not discussed in this paper in further detail.

3.2.3. Numerical-experimental validation

To support the use of the present methodology, the CFD model has been previously validated against dedicated experimental data for both the scaled ICE2 train and the full-scale ETR1000 configuration.

For the ICE2 train, tested in the original-length TRAIN Rig tunnel, the numerical-experimental comparison is presented and discussed in detail in Negri et al. (2025b). The validation considers the normalized longitudinal velocity component U_x^{norm} , comparing the CFD slipstream history with the ensemble statistics from Gilbert et al. (2013) at the reference measurement point C1. The results (Fig. 12(a)) show that the model accurately reproduces the piston-wind development and the nose-induced velocity peak, while discrepancies in the reverse-flow region are consistent with the well-known limitations of Cobra probes in detecting backflow. A good agreement is also observed in the wake region, where the CFD peak magnitude and its position fall within the experimental variability.

For the full-scale ETR1000 case, the validation against measurements collected in the Turin Rebaudengo tunnel is presented in Negri et al. (2024, 2025a). The corresponding CFD sampling point is located at a height of 1.2 m above the platform ground and 2.5 m laterally from the track center, matching the experimental probe position. As illustrated in Fig. 12(b), the CFD simulation correctly captures the nose peak, the reverse flow along the train body, and the overall wake evolution, remaining within the experimental envelope defined by the ensemble mean and standard deviation. Differences near the tunnel wall and in the highly turbulent wake are in line with the variability observed between repeated experimental passages and with

the expected sensitivity to local boundary-layer dynamics. In particular, since the CFD simulation provides a single deterministic realization, whereas the experiments supply an ensemble mean with its standard deviation, discrepancies in the wake region mainly reflect the intrinsic statistical variability of the slipstream rather than deficiencies of the numerical model.

In the present paper, these validation results are recalled only to establish the reliability of the underlying CFD model employed in the simplified framework, while the reader is referred to the referenced works for a complete interpretation of the results, including detailed probe positioning and limitations, flow-physics discussion, and sensitivity to tunnel geometry.

3.2.4. Model verification: comparison against moving mesh simulation

The simplified steady-domain approach with fixed mesh (hereafter referred to as MF) is verified by comparing its slipstream predictions against unsteady moving-mesh simulations (MM). Two configurations are considered: the ETR1000 train operating in the uniform-section tunnel, and the scaled-model ICE2 train in a 1:25 tunnel environment of extended length of 24 m. For each MM case, an ensemble-based slipstream statistic is constructed by sampling at specific probe locations, applying a minimum longitudinal separation of 20 m (full-scale or in equivalent scaled dimensions) to ensure statistical independence, using 9 probes for the ETR1000 configuration and 20 probes for the ICE2 configuration. The resulting ensemble average and standard deviation are compared with the time-averaged quantities extracted from the MF simulations. In the MF approach, the time-averaged fields are computed over a temporal window equivalent to the travel time of the train across the longitudinal interval spanned by the MM probes used for the ensemble statistics, ensuring consistency between the two averaging procedures.

For the MF simulations, the inlet boundary condition is set equal to the relative velocity given by the local piston-wind speed sampled when the train nose reaches a reference position located in tunnel, sufficiently downstream from the tunnel entrance to ensure that the train is fully immersed in the confined environment and that the piston wind has reached a quasi-stationary state. In the context of the present study, this reference location corresponds to the tunnels mid-point, i.e. approximately 300 m from the tunnel entrance for the ETR1000 and 12 m for the ICE2 configuration. Although the piston wind increases modestly as the MM train continues to advance, this effect is not present under MF conditions and its effect will be highlighted in the results discussion.

The slipstream velocity for the models comparison is evaluated at fixed probe locations. For the ETR1000 case, four measurement points are considered to assess the robustness of the methodology across different flow regions, located at 2.5 and 3.0 m (TSI position) from the center of the rails (COR) in the lateral direction, and at

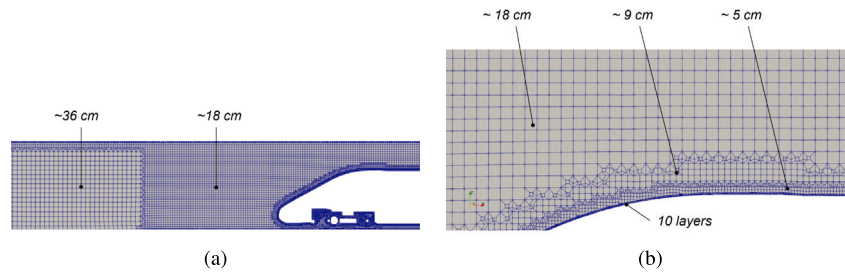


Fig. 11. Visualization of ETR1000 mesh density and refinement regions.

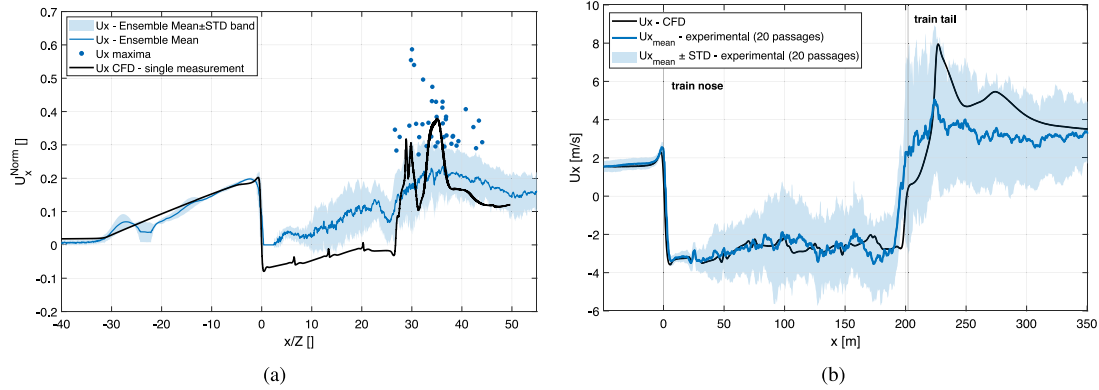


Fig. 12. Numerical-experimental validation plots for ICE2 in TRAIN Rig laboratory (a) and ETR1000 train in Turin Rebaudengo tunnel (b).

different heights above the top of the rail (TOR): $z = 1.4$ m and $z = 1.75$ m, approximately at mid-height of the train body. For the ICE2 configuration, the slipstream is sampled at the reference position C1, as defined in Gilbert et al. (2013) and Negri et al. (2025b).

Fig. 13 presents the slipstream predictions obtained for the ETR1000 configuration. The MF model reproduces the velocity jump at the nose ($X = 0$ m) with good consistency relative to the MM reference, at all the measurement positions. This behavior is expected, as the flow around the nose is predominantly inviscid and the velocity reversal is primarily governed by the blockage ratio between train and tunnel, given a specific train speed and piston wind condition. Along the body region, the development of the boundary layer is captured with comparable trends between the MM and MF methodology. At the vertical position of 1.4 m, the MF model exhibits an underestimation of the mean velocity, i.e. more negative values, with a maximum deviation of approximately 1.6 m/s at the 2.5 m lateral position (Fig. 13(a)) and about 0.8 m/s at the 3.0 m position (Fig. 13(c)). For the same lateral positions at $Z = 1.75$ m, the deviations are smaller and remain below 0.8 m/s in both cases. The velocity discrepancies can be related to differences in the flow development along the rear portion of the train, where the MM solution exhibits a stronger wall-driven acceleration and a more pronounced near-wall momentum deficit, i.e. thicker boundary layer, which results in higher velocity magnitudes. This may be attributed to the gradual increase of piston wind present in MM but absent in MF due to the fixed velocity boundary condition. According to continuity considerations, an increase in piston wind speed results in a less intense opposing flow around the train, leading to reduced confinement of the train boundary layer. In addition, a secondary contribution may arise from the presence of the ACMI interface in the MM setup: the interface inherently introduces a local modification of the mesh and, given the reduced thickness of the train boundary layer, such local mesh alterations may affect its development in specific regions. In the tail and wake regions, the velocity sudden increase ($X = 202$ m) is comparable between the two methodologies at all measurement positions, confirming that the wake development is consistently reproduced. The remaining discrepancies, including the under-prediction of

the near-wake peak in MF, stem from the velocity differences already present in the boundary-layer region along the train body, which are convected downstream and appear in the wake region as well, where they manifest as a reduced peak intensity in the MF results. Finally, the standard deviation bands in the train and wake regions have comparable magnitude and they develop at the same location along the train itself, approximately after the third coach, indicating that the turbulence-driven variability of the annular flow and wake is consistently represented by the fixed-mesh approach.

A similar level of agreement is observed for the scaled ICE2 configuration results, in Fig. 14. The velocity jump at the nose ($X = 0$ m) is again accurately reproduced, as it primarily depends on blockage and boundary-condition consistency. In the body region, the boundary-layer profile matches between the two approaches, with the maximum absolute deviation remaining within 0.5 m/s, indicating that the boundary-layer growth and its underlying physics are well captured also under MF conditions. At the tail, about $X = 4.2$ m, the velocity sudden increase is captured with good agreement, while the near-wake peak is under-predicted by approximately 1.3 m/s in MF, a difference attributable to the continuous temporal averaging, which inherently smooths short velocity peaks that could be preserved in the ensemble-based MM statistics. Downstream, the far-wake decay occurs more rapidly in the MM results, likely due to the higher velocity peak near the train tail, which produces a larger deviation from the undisturbed flow and thus a faster recovery toward the ambient velocity compared with the more gradual decay observed in MF. The MF standard deviation band, reflecting the cumulative temporal variability of the flow in this region, is comparable between the tested methodologies.

3.2.5. Computational efficiency and limitations

One objective of the simplified steady-domain model is to reduce the computational cost associated with simulating the near-field flow around a train, compared with a full moving-mesh simulation. A first significant improvement is already achieved by modeling the piston wind through 2D CFD simulations, which avoids the need to resolve

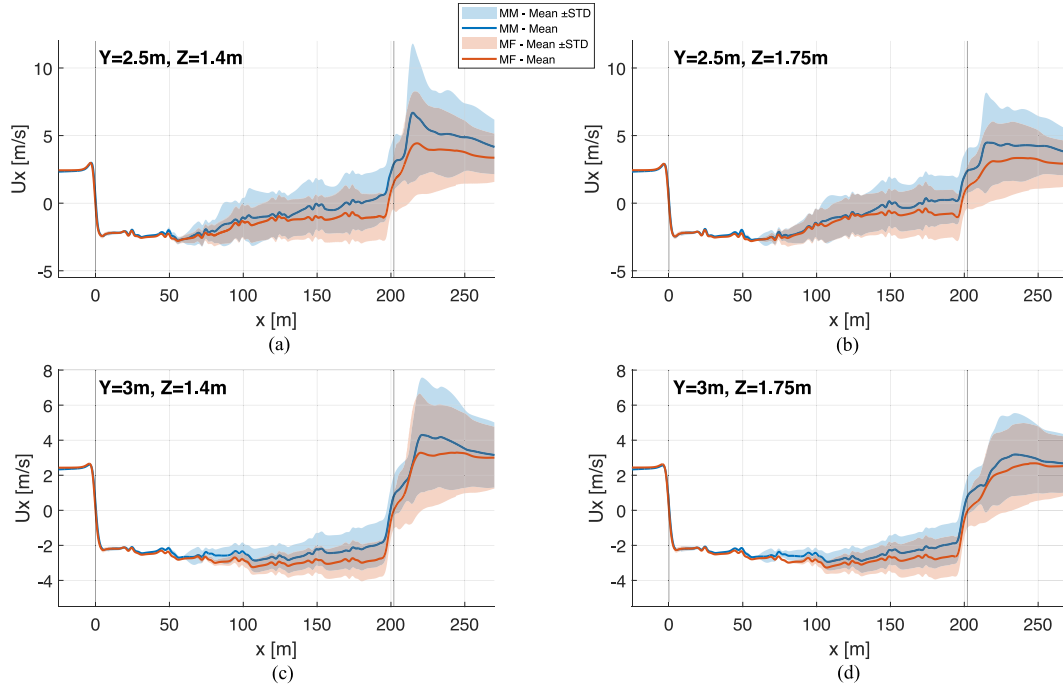


Fig. 13. Slipstream longitudinal velocity profiles for ETR1000 at 4 measurement positions: solid lines show the mean, shaded bands denote mean $\pm$ standard deviation. Flow measurement positions: (a) $Y = 2.5$ m, $Z = 1.4$ m; (b) $Y = 2.5$ m, $Z = 1.75$ m; (c) $Y = 3.0$ m, $Z = 1.4$ m; (d) $Y = 3.0$ m, $Z = 1.75$ m.

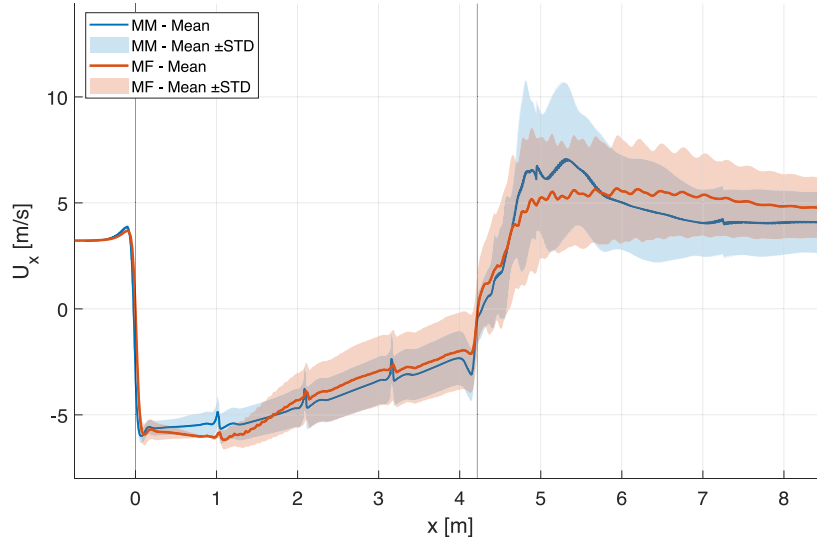


Fig. 14. Comparison of slipstream longitudinal velocity for ICE2 train between the moving-mesh (MM) simulation and the fixed-mesh (MF) approach. Mean profiles are shown as solid lines, with shaded bands denoting $\pm$ standard deviation.

the complete entrance transient. Building on this aspect, further computational gains are obtained by computing the train near-field aerodynamics on a fixed mesh, eliminating the interpolation overhead associated with the mesh interfaces and allowing a reduction of mesh density in regions of minor interest, such as the open-air domain which are not represented in the MF approach. Referring to the full-scale ETR1000 case, the ratio between the number of cells in MF and MM is approximately 1.17; the MF number of elements remains relatively large because, other than the train mesh itself, sufficient upstream and downstream extents are still required to capture wake development. In terms of computational cost, the speed-up factor computed as the ratio

of core-hours between MM and MF amounts to 2.75, meaning that MF is almost three times faster than MM under the tested conditions.

Despite the advantages, the limitations of the MF approach must be acknowledged. First, the method requires the tunnel geometry to be uniform along the longitudinal direction, so that a quasi-steady flow state can be represented. This assumption is valid when both the tunnel and platform regions are sufficiently long relative to the train dimensions and are not affected by local geometric discontinuities. Second, in the present formulation, the inlet boundary condition is imposed using a constant piston-wind value sampled at a reference location. Experimental and numerical results indicate that the piston wind exhibits a transient evolution even after the train is fully inside

the tunnel. The behavior is particularly evident in tunnels of moderate length, such as those considered in the present study, where the transient has not yet fully decayed and the piston-wind magnitude continues to increase slightly as the train advances; this effect is not currently captured in MF. A possible improvement of the model would be to impose a time-varying inlet velocity, replicating the full piston-wind time history. Finally, a moderate underestimation of the train second half of the boundary layer region, and consequently of the near-wake peak velocity, is observed in MF. This effect may be reduced by introducing a time-dependent piston-wind profile and quantified through further studies. Nevertheless, the MF approach reproduces the key slipstream features of both configurations with good fidelity, including the velocity reversal at the nose, the boundary-layer growth, and the near-wake peak, remaining suitable for sensitivity analyses and comparative assessments, where relative differences between operating conditions or geometries are of primary interest and where statistical consistency of the slipstream behavior is relevant.

4. Case studies

A set of representative tunnel–train case studies is introduced to verify the capability of the proposed methodology to distinguish between different configurations. Variations in blockage ratio and measurement location are considered to assess the model behavior under changing conditions, while also providing an example of the model application for its intended engineering use, without the intent of exploring the entire parameter space. The study focuses on rectangular tunnel sections, chosen to remain consistent with the full-scale ETR1000 validation case while avoiding local geometric features such as platforms or wall recesses. This choice ensures that the analysis isolates the effect of global blockage on the confined aerodynamics, without introducing site-specific disturbances. The proposed framework is employed to investigate how different moderate blockage ratios influence the slipstream and wake development around a high-speed train. The operating conditions are selected to remain realistic for scenarios involving tunnels equipped with platforms. From literature (Negri et al., 2023, Negri et al., 2025c) and railway line regulations, trains generally operate at reduced speeds when approaching or passing platforms: accordingly, a train speed of 120 km/h is adopted, allowing the study to capture plausible near-field aerodynamics without venturing into extreme operational regimes.

Four tunnel configurations are considered; the geometry of each case is illustrated in Fig. 15, while the main geometric dimensions, train-tunnel blockage ratio and the estimated piston-wind velocity obtained from the 2D model are reported in Table 2. Slipstream velocities are sampled at six positions, combining two lateral offsets and three vertical elevations. The lateral distances, $Y=2.5$ m and $Y=3.0$ m from the center of rails (COR), reflect typical safety-relevant locations: the TSI reference distance of 3.0 m, and the 2.5 m offset, representative of the yellow safety line adopted in Italy and in several international environments. Vertically, measurements are taken at $Z=1.05$, 1.40, and 1.75 m from the top of rail (TOR), providing sensitivity to variations in slipstream intensity near the lower and mid regions of the train body. All measurement locations are depicted in Fig. 15, while the slipstream longitudinal velocities obtained from the simulations are shown in Fig. 16. For all configurations and measurement locations, the slipstream analysis considers both the mean, i.e. time average, and the mean $\pm$ standard deviation of the longitudinal velocity component. In the following figures, the solid line represents the time-averaged velocity, while the shaded bands denote the time-averaged value plus or minus one standard deviation, thereby capturing the flow's stochastic variability and the expected range of peak velocities.

Table 2

Tunnels characteristics and piston-wind estimations for each case study.

Case	Tunnel section (w $\times$ h) [m]	β	Piston wind (2D) [m/s]
P1	14.68 $\times$ 6.29	0.119	2.20
Base	14.00 $\times$ 6.00	0.131	2.40
M1	13.28 $\times$ 5.69	0.146	2.80
M2	9.78 $\times$ 5.50	0.205	3.82

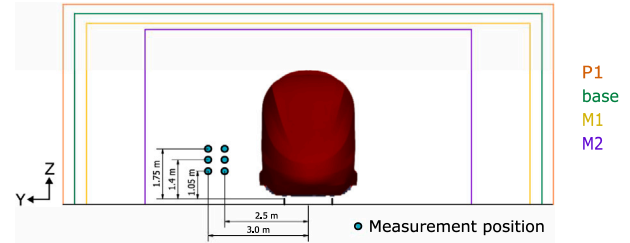


Fig. 15. Tunnel characteristics and measurement locations for the P1, base, M1 and M2 case studies.

4.1. Influence of blockage and measurement position

At the TSI reference position ($Y = 3.0$ m from COR, $Z = 1.40$ m from TOR), the influence of tunnel blockage is shown in Fig. 16(d). The magnitude of the velocity jump at the train nose ($X = 0$ m), increases systematically with blockage: comparing the lowest-blockage case P1 with the highest M2, the peak-to-peak variation across the nose is approximately 45% larger, evidencing a stronger confined acceleration effect. Along the train body, the magnitude of the reverse flow also grows with blockage: case M2 produces intense adverse velocities exceeding -5 m/s, while in the other cases (P1, base, M1) the backflow is weaker and gradually stabilizes as the train progresses, reflecting the interaction between the developing train boundary layer and the annular reverse stream. For M2, instead, the reverse flow remains dominant over the entire train length, suggesting that the boundary-layer contribution is largely suppressed. At the tail, all configurations show a rapid velocity rise associated with wake formation. Although the mean near-wake average velocity maximum remains comparable across blockages (around 3 m/s), the inclusion of the standard deviation, as assessment of the velocity peak, highlights sensible differences: the highest-blockage case M2 reaches approximately 6.8 m/s, about 1.27 times the corresponding value for P1. Further downstream, in the far wake region, the velocity decays toward the pre-existing piston-wind level imposed at the inlet for each scenario. The same qualitative trends are observed at the other measurement positions, both laterally ($Y = 2.5$ m and 3.0 m from COR) and vertically ($Z = 1.05$, 1.40, 1.75 m from TOR), with differences in magnitude that reflect the local balance between the boundary-layer thickness and the confined annular flow, as well as the influence of the measurement location on the wake intensity.

Comparing the two lateral measurement locations at 2.5 m and 3.0 m from the COR, two main effects emerge. Firstly, the influence of the train boundary layer is measured earlier at 2.5 m: between nose and tail, the longitudinal velocity increases, i.e. becoming less negative, already from the third coach, around 80 m after train nose, especially at the lower vertical positions, where the flow is strongly affected by bogies and underbody regions that are more exposed and highly turbulent for a high-speed train. Secondly, in the wake, moving away from the train markedly reduces the peak intensity: for the base blockage case at $Z = 1.40$ m, the near-wake peak defined as mean plus standard deviation at 3.0 m is about 19.5% lower than at 2.5 m (Figs. 16(c) and (d)). This attenuation can be attributed to the lateral spreading and progressive dissipation of the wake vortices as they convect downstream from the tail and it is consistent with full-scale

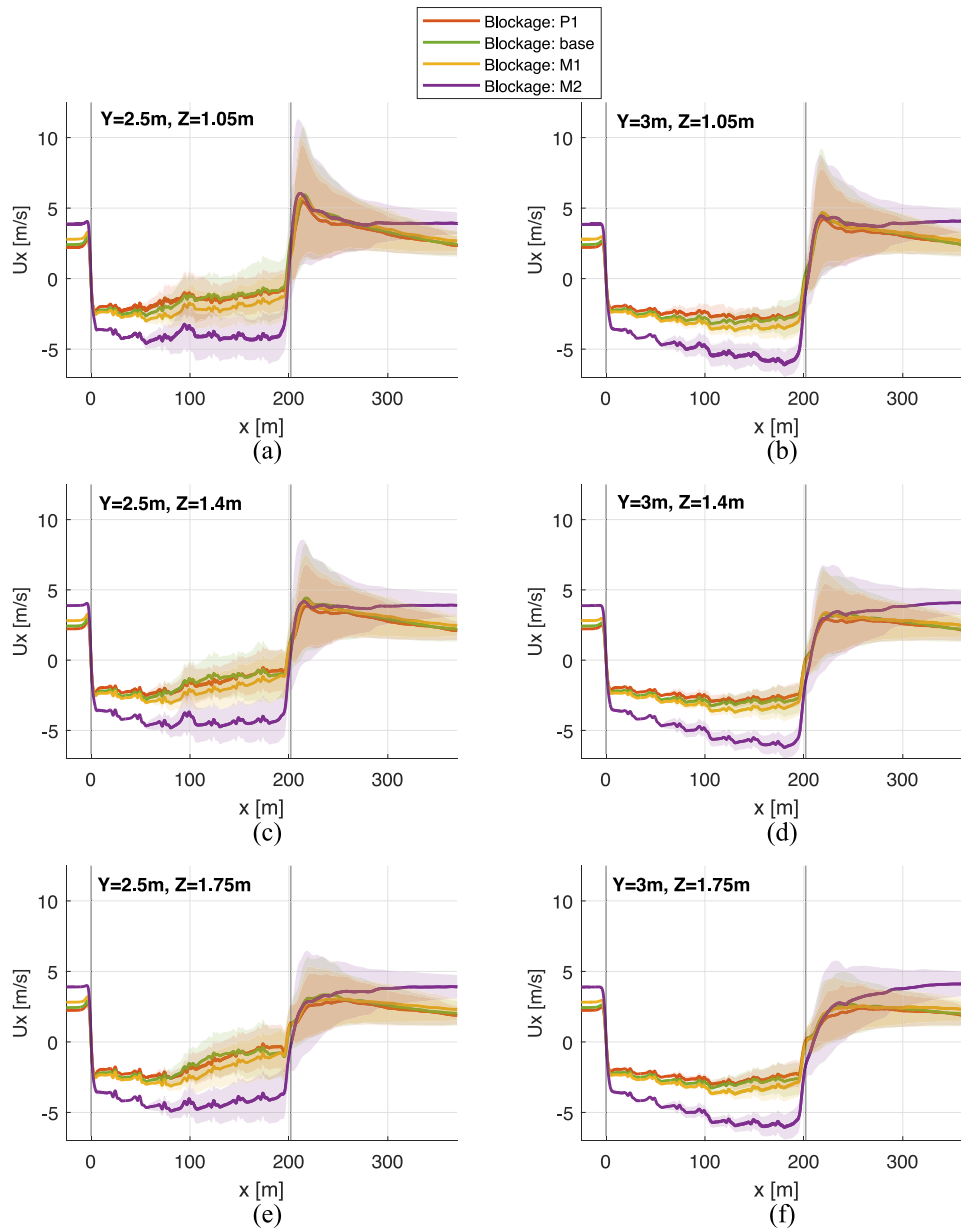


Fig. 16. Slipstream longitudinal velocity profiles for P1, base, M1 and M2 blockage cases: solid lines show the mean, shaded bands denote mean $\pm$ standard deviation. Flow measurement positions: (a) $Y = 2.5\text{ m}$, $Z = 1.05\text{ m}$; (b) $Y = 3\text{ m}$, $Z = 1.05\text{ m}$; (c) $Y = 2.5\text{ m}$, $Z = 1.4\text{ m}$; (d) $Y = 3\text{ m}$, $Z = 1.4\text{ m}$; (e) $Y = 2.5\text{ m}$, $Z = 1.75\text{ m}$; (f) $Y = 3\text{ m}$, $Z = 1.75\text{ m}$.

measurements reported in Negri et al. (2023), where similar differences between 2.5 m and 3.0 m were observed in terms of boundary-layer growth and reduction of wake peaks under comparable blockage and train speed conditions.

With respect to vertical measurement positions, focusing on the base blockage at $Y = 2.5\text{ m}$ distance from COR (Figs. 16(a), (c) and (e)), the three measurement heights confirm that the most critical slipstream exposure occurs closer to the ground. The wake peak (mean plus standard deviation) decreases by about 23% when moving from $Z = 1.05\text{ m}$ to 1.40 m , and by roughly 27% when moving from $Z = 1.4\text{ m}$ to 1.75 m . Similar vertical trends are observed for the other blockage configurations and at $Y = 3.0\text{ m}$. Overall, for a given lateral offset, the lowest sampling height systematically exhibits the largest wake peaks, indicating that the region closer to the underbody and bogies remains the most demanding from an aerodynamic safety perspective. This behavior is consistent with open-air measurements (Baker et al., 2014), which show that the highest slipstream intensities occur at the lowest

sampling heights, indicating that the trend is governed by the train-induced flow rather than by confinement. Similar vertical distributions are also reported in tunnel studies as Li et al. (2021) and Li et al. (2020), supporting the interpretation derived from the present analyses.

4.2. Interpretation of results

Several observations emerge from the case-study analysis. The variations induced by changing the blockage ratio confirms that the proposed framework is sufficiently sensitive to capture meaningful aerodynamic differences. In the case studies analyzed, these differences manifest primarily in the boundary-layer region: changes in blockage modify the intensity of the annular counter-flow and its interaction with the train boundary layer, producing the most pronounced variations between nose and tail. In the wake region, variations are still present but remain comparatively limited, with mean wake velocities showing only modest differences across the four blockage cases. The

findings suggest that for the moderate blockage ratios and reduced train speeds considered here, the effects of confinement are mainly limited to the pre-wake region, while the near-wake topology and its associated velocity peaks remain broadly similar to those observed for an unconfined train. This behavior is physically consistent and aligns with observations from full-scale measurements in Negri et al. (2025c), which represents comparable train and tunnel conditions to those adopted in this work, where slipstream velocities in open air and in tunnel sections without geometrical variations displayed nearly identical wake velocities, while substantial differences were found ahead of the tail due to piston wind and annular reverse flow. Nevertheless, it should be noted that literature, such as Li et al. (2021), indicates that for higher ratios or higher operational speeds, wake characteristics may become far more sensitive to tunnel geometry, implying that the trends observed in this study apply to the present mild operating regime and should not be extrapolated to more extreme configurations.

5. Conclusions

This study presented a novel and computationally efficient methodology for reproducing the aerodynamic slipstream in railway tunnels, based on the decoupling between the piston-wind contribution and the near-field flow around the train. The methodology was validated using high-speed trains in both full-scale and reduced-scale configurations, demonstrating that it provides a physically consistent and efficient alternative to conventional fully unsteady moving-mesh simulations. Finally, the methodology was applied to a set of case studies involving four tunnel blockage ratios, demonstrating its capability to resolve physically meaningful aerodynamic differences between configurations and to assess the influence of tunnel cross-section and measurement position.

The main findings can be summarized as follows:

- The two-dimensional CFD model accurately reproduces the piston-wind evolution for both the full-scale ETR1000 and the 1:25 scaled ICE2 train configurations. The simplified 2D approach captures the amplitude and shape of the longitudinal velocity profiles with deviations below 3%–8% compared with the 3D and experimental references, while requiring only a fraction of the computational cost, achieving a speed-up exceeding two orders of magnitude.
- The three-dimensional steady-domain model reproduces the slipstream statistics of the reference moving-mesh simulations with good accuracy. The fixed-mesh results capture the velocity drop around the nose, the boundary-layer development along the train, and the wake velocity peak. The differences between the two approaches are mainly observed in the boundary-layer and near-wake regions, where the mean velocities tend to be underestimated.
- The standard-deviation envelopes predicted by the fixed-mesh model remain within the variability range of the moving-mesh ensemble, confirming that the proposed approach is capable of reproducing the stochastic features of the confined turbulent flow.
- In terms of computational cost, the 2D moving-mesh model yields a speed-up of over two orders of magnitude compared with the corresponding 3D piston-wind simulation, while the 3D fixed-mesh steady model is roughly 2.75 times faster than the full moving-mesh simulation thanks to the absence of dynamic mesh motion, ACMI interpolation and the reduced far-field domain.
- The case studies assessment, concerning difference blockage ratios, confirmed that the methodology resolves the aerodynamic differences associated with the four blockages ranging from 0.1 to 0.2. In this conditions, the strongest slipstream modifications appear in the boundary-layer region, while wake variations remain modest. The influence of measurement position was also correctly reproduced, with lower heights and smaller lateral offsets yielding higher slipstream peaks, consistent with open-air and tunnel literature.

Finally, the proposed methodology can efficiently simulate a wide range of train and tunnel configurations, enabling the creation of aerodynamic reference maps useful for sensitivity analyses and comparative assessments across different geometries and operating conditions. Overall, the coupled CFD framework offers a physically consistent and computationally efficient approach for the analysis of aerodynamic flows in confined railway environments, suitable for large-scale parametric studies, design optimization, and early-stage evaluation of tunnel–train interactions.

Despite its efficiency and accuracy, the proposed approach presents some limitations. Its application requires a tunnel with a geometrically uniform cross-section, and it assumes a constant piston-wind condition as prescribed inlet velocity, whereas in reality the piston velocity could vary as the train moves through the tunnel. Future developments could address this limitation by introducing a time-dependent inlet boundary condition driven by the piston-wind prediction. Further extensions will consider different tunnel configurations, including the presence of passenger platforms, as well as the inclusion of gas compressibility effects, which are expected to become relevant at higher blockage ratios, higher train speeds, and for longer train configurations; in such cases, compressible flow formulations will be adopted to account for pressure-wave propagation and local flow acceleration in the train–tunnel annular region.

CRediT authorship contribution statement

S. Negri: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **G. Tomasini:** Writing – review & editing, Visualization, Supervision, Investigation, Conceptualization. **P. Schito:** Visualization, Software, Resources, Methodology, Investigation. **D. Rocchi:** Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Araya Reyes, C.E., Brambilla, E., Schito, P., Tomasini, G., et al., 2022. Safety to crosswind of railway vehicles passing by a windbreak gap: numerical-experimental methodology. In: *Railways* 2022. pp. 5–5.
- Baker, C.J., Quinn, A., Sima, M., Hoefener, L., Licciardello, R., 2014. Full-scale measurement and analysis of train slipstreams and wakes. part 1: Ensemble averages. *Proc. Inst. Mech. Eng. Part F: J. Rail Rapid Transit* 228 (5), 451–467.
- Bowman, J., 2015. It was a tragic accident: Toddler killed when pram sucked onto tracks by high-speed train. *Mirror* Published on 29 April 2015.
- Brambilla, E., Araya Reyes, C.E., Schito, P., Tomasini, G., et al., 2021. Procedure for evaluating crosswind safety of a train running in presence of windbreaks with gap, part 1: determination of the aerodynamic loads. In: *Proceedings of Aerovehicles* 4. pp. 152–155.
- Brambilla, E., Schito, P., Somaschini, C., Rocchi, D., 2022. Virtual homologation of high-speed trains in railway tunnels: A new iterative numerical approach for train-tunnel pressure signature. *Proc. Inst. Mech. Eng. Part F: J. Rail Rapid Transit* 236 (5), 511–531.
- Du, J., Fang, Q., Zhang, X., Wang, H., 2024. Aerodynamic effect associated with tunnel length. *KSCE J. Civ. Eng.* 28 (7), 2997–3008.
- Fu, M., Li, P., Liang, X., 2017. Numerical analysis of the slipstream development around a high-speed train in a double-track tunnel. *Plos One* 12 (3), e0175044.
- Gilbert, T., Baker, C., Quinn, A., 2013. Gusts caused by high-speed trains in confined spaces and tunnels. *J. Wind Eng. Ind. Aerodyn.* 121, 39–48.
- Hemida, H., 2023. Contribution of computational wind engineering in train aerodynamics—past and future. *J. Wind Eng. Ind. Aerodyn.* 234, 105352.

- Iliadis, P., Hemida, H., Soper, D., Baker, C., 2020. Numerical simulations of the separated flow around a freight train passing through a tunnel using the sliding mesh technique. *Proc. IMechE Part F: J. Rail Rapid Transit* 234 (6), 638–654.
- Jiang, Z., Liu, T., Chen, X., Li, W., Guo, Z., Niu, J., 2019. Numerical prediction of the slipstream caused by the trains with different marshalling forms entering a tunnel. *J. Wind Eng. Ind. Aerodyn.* 189, 276–288.
- Khayrullina, A., Blocken, B., Janssen, W., Straathof, J., 2015. CFD simulation of train aerodynamics: train-induced wind conditions at an underground railroad passenger platform. *J. Wind Eng. Ind. Aerodyn.* 139, 100–110.
- Li, W., Gu, Y., Deng, Y., Zhao, W., Fan, X., 2025a. Aerodynamic behaviors on a streamlined-passenger train and an express-freight train meeting in a same tunnel. *Wind. Struct.* 40 (2), 75–92.
- Li, W., Gu, Y., Liu, T., Deng, Y., Zhao, W., Fan, X., 2025b. Mitigating aerodynamic effects through segment length allocation of variable cross-section tunnels. *Tunn. Undergr. Space Technol.* 157, 106341.
- Li, W., Gu, Y., Zhao, W., Deng, Y., Fan, X., 2025c. Alleviating tunnel aerodynamics through hybrid suction & blowing techniques applied to train nose sections. *J. Wind Eng. Ind. Aerodyn.* 256, 105961.
- Li, W., Liu, T., Chen, Z., Guo, Z., Huo, X., 2020. Comparative study on the unsteady slipstream induced by a single train and two trains passing each other in a tunnel. *J. Wind Eng. Ind. Aerodyn.* 198, 104095.
- Li, W., Liu, T., Martinez-Vazquez, P., Chen, Z., Huo, X., Guo, Z., 2021. Influence of blockage ratio on slipstreams in a high-speed railway tunnel. *Tunn. Undergr. Space Technol.* 115, 104055.
- Liu, M., Zhang, H., Zhu, C., Zheng, W., You, S., 2020. Theoretical modeling of piston wind induced by multiple trains in longitudinal tunnel. *Sustain. Cities Soc.* 57, 102127.
- Liu, Z., Zhou, D., Soper, D., Chen, G., Hemida, H., Guo, Z., Li, X., 2023. Numerical investigation of the slipstream characteristics of a maglev train in a tunnel. *Proc. IMechE Part F: J. Rail Rapid Transit* 237 (2), 179–192.
- Liu, M., Zhu, C., Cui, T., Zhang, H., Zheng, W., You, S., 2018. An alternative algorithm of tunnel piston effect by replacing three-dimensional model with two-dimensional model. *Build. Environ.* 128, 55–67.
- Meng, S., Zhou, D., Wang, Z., 2019. Moving model analysis on the transient pressure and slipstream caused by a metro train passing through a tunnel. *Plos One* 14 (9), e0222151.
- Menter, F., 1993. Zonal two equation kw turbulence models for aerodynamic flows. In: 23rd Fluid Dynamics, Plasmadynamics, and Lasers Conference. p. 2906.
- Menter, F.R., 1994. Two-equation eddy-viscosity turbulence models for engineering applications. *AIAA J.* 32 (8), 1598–1605.
- Negri, S., Tomasini, G., Araya Reyes, C.E., Rocchi, D., 2025a. CFD analysis of flow patterns induced by high-speed and freight trains in tunnels and their impact on platform regions. *Int. J. Rail Transp.* Submitted for publication.
- Negri, S., Tomasini, G., Rocchi, D., Schito, P., Soper, D., Hemida, H., 2025b. Numerical assessment of train slipstream in tunnels: Stochastic analysis from CFD data. *J. Wind Eng. Ind. Aerodyn.* 264, 106130.
- Negri, S., Tomasini, G., Schito, P., Rocchi, D., 2023. Full scale experimental tests to evaluate the train slipstream in tunnels. *J. Wind Eng. Ind. Aerodyn.* 240, 105514.
- Negri, S., Tomasini, G., Schito, P., Rocchi, D., Semeraro, F.F., Araya Reyes, C.E., 2024. Experimental-numerical analysis of high-speed train slipstream in open air and confined spaces. In: Pombo, J. (Ed.), *Proceedings of the Sixth International Conference on Railway Technology: Research, Development and Maintenance*, Civil-Comp Press, Edinburgh, UK, Online Volume: CCC 7, Paper 3.14.
- Negri, S., Tomasini, G., Schito, P., Semeraro, F.F., Rocchi, D., 2025c. Train slipstream assessment in open air and confined spaces through full-scale experimental tests. *Proc. Inst. Mech. Eng. Part F: J. Rail Rapid Transit* 09544097251364414.
- Ouyang, M., Chen, S., Li, Q., Sun, Y., 2025. Bottom flow patterns of high-speed trains in infinite long tunnel. *Phys. Fluids* 37 (3).
- Pan, S., Fan, L., Liu, J., Xie, J., Sun, Y., Cui, N., Zhang, L., Zheng, B., 2013. A review of the piston effect in subway stations. *Adv. Mech. Eng.* 5, 950205.
- RAIB, 2017a. Occupied Wheelchair Contacting Passing Train at Twyford Station. Technical Report, UK Department for Transport, RAIB Report 03/2017. (Accessed 03 November 2025).
- RAIB, 2017b. Report R012017: Near Miss Involving a Signaller at Twyford Station. Technical Report, UK Department for Transport, (Accessed 03 November 2025).
- Semeraro, F.F., Cioffi, A., Pellegrino, E., Schito, P., Vignati, M., 2022. Numerical analysis of wind-break fences for truck stability in crosswind. *SAE Int. J. Commer. Veh.* 16 (02-16-02-0011), 165–177.
- Standard, C., 2009a. Railway applications—aerodynamics—Part 4: Requirements and test procedures for aerodynamics on open track. CEN EN14067-4.
- Standard, C., 2009b. Railway applications—aerodynamics—Part 5: Requirements and test procedures for aerodynamics in tunnels. CEN EN14067-5.
- Tages Anzeiger, 2015. Unattended pushchair drawn into the path of a freight train: A tragic incident at a station platform. *Tages Anz.*
- T.S.I. LOC&PAS, 2014. Technical specification for interoperability (TSI) 2014: Locomotives and passenger rolling stock (loc&pas). (Accessed 13 January 2025), <https://eur-lex.europa.eu/eli/reg/2014/1302/oj>.
- Xiong, X., Cong, R., Li, X., Geng, Y., Tang, M., Zhou, S., Na, Y., Jiang, C., 2022. Unsteady slipstream of a train passing through a high-speed railway tunnel with a cave. *Transp. Saf. Environ.* 4 (4), tdac032.
- Yan, X., Tao, L., Peng, J., Zeng, Y., Fang, Y., Bai, Y., 2020. Behavior of piston wind induced by braking train in a tunnel. *Energies* 13 (23), 6420.